Stage 5 Report: Final Evaluation and Analysis

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 3 Nov 2026 Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

1. Completed experiments

Four pretrained methods ran on every input: bicubic, Real-ESRNet, Real-ESRGAN and ResShift v3 (4 steps). Nothing is trained. The test split has 276 images (80 objects, 100 faces, 96 text), downsampled ×4 to 128×128 and compressed at JPEG none, 50, 30 and 10: 1,104 inputs per method. Dev has 64 images.

- **Hardware.** Tesla T4, batch size 1. Real-ESRNet and Real-ESRGAN use float32; ResShift uses float32 weights with float16 autocast. Bicubic runs on the notebook CPU. Metrics ran on an Apple M4 CPU.
- **Frozen settings.** `configs/final.yaml` (SHA-256 `70f6c341`) was frozen before the test run from dev HR images only. OCR: Tesseract 5.5.1, `--psm 4 --oem 1`, the best of 18 candidates, HR CER 0.017 on dev. ArcFace: buffalo_l with a 512 px detector input (640 px missed 12 of 20 dev faces); match threshold 0.182, the 1% false match rate on 190 dev impostor pairs. MUSIQ: KonIQ-10k checkpoint. ResShift: seed offsets 0 to 4. Intervals: 95% percentile bootstrap [lo, hi], 10,000 resamples, seed 42, no multiple-comparison correction. The analysis and failure-selection rules are in the same file.
- **Reproduction.** With Pillow's BASIC text layout, 320 of 320 dev inputs equal the stage 4 archive. Bicubic, Real-ESRNet and Real-ESRGAN reproduce the archived per-image metrics within tolerance: at most 0.0049 dB PSNR, 0.0002 SSIM and 0.0005 LPIPS. ResShift does not: at most 0.032 dB PSNR (18 of 256 values above 0.01 dB) and 0.0013 LPIPS; the mean absolute PSNR difference is 0.0035 dB. Inputs, weights and code are identical by SHA-256. Stage 4 used CUDA 13.0 and this run CUDA 12.8, which probably explains the difference.
- **Test access and reruns.** An exploratory test run exists from commit `cc9a873`; no setting used it. The test notebook ran twice: the first outputs were lost in a Kaggle restart before any metric was computed. After the test evaluation, one bug was fixed: ArcFace chose the largest face by its unclipped box, so a partial face at the border became the reference in 2 test images. Boxes are now clipped to the image. No dev face and no frozen setting changed; only the metrics were recomputed. Tables before the fix: `results/final/arcface_unclipped/`.

2. Final results (test split, 276 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet; MUSIQ is no-reference. Last 2 columns: CER for text; ArcFace similarity to the HR face and match rate for faces. Means over images; best per content and condition in bold.

No JPEG

Content	Method	PSNR ↑	SSIM ↑	LPIPS ↓	MUSIQ ↑	CER ↓ / ArcFace sim ↑	Match rate ↑
Objects	Bicubic	26.69	0.718	0.465	33.1	n/a	n/a
Objects	Real-ESRNet	**27.54**	**0.765**	0.357	58.1	n/a	n/a
Objects	Real-ESRGAN	25.65	0.709	0.244	**66.4**	n/a	n/a
Objects	ResShift	26.56	0.731	**0.192**	65.7	n/a	n/a
Objects	HR reference	n/a	n/a	n/a	68.6	n/a	n/a
Faces	Bicubic	32.08	0.869	0.296	41.5	**0.983**	**1.00**
Faces	Real-ESRNet	**32.68**	**0.886**	0.245	62.3	0.977	1.00
Faces	Real-ESRGAN	30.25	0.852	0.166	71.5	0.954	1.00
Faces	ResShift	31.32	0.858	**0.113**	**72.4**	0.972	1.00
Faces	HR reference	n/a	n/a	n/a	73.6	n/a	n/a
Text	Bicubic	16.42	0.706	0.436	35.5	**0.210**	n/a
Text	Real-ESRNet	19.25	0.838	0.184	49.4	0.220	n/a
Text	Real-ESRGAN	17.79	0.819	0.107	**55.0**	0.340	n/a
Text	ResShift	**19.72**	**0.859**	**0.086**	54.0	0.243	n/a
Text	HR reference	n/a	n/a	n/a	57.4	0.019	n/a

JPEG 10

Content	Method	PSNR ↑	SSIM ↑	LPIPS ↓	MUSIQ ↑	CER ↓ / ArcFace sim ↑	Match rate ↑
Objects	Bicubic	23.52	0.570	0.758	19.6	n/a	n/a
Objects	Real-ESRNet	**23.85**	**0.609**	0.595	40.7	n/a	n/a
Objects	Real-ESRGAN	22.67	0.571	0.440	**60.5**	n/a	n/a
Objects	ResShift	22.98	0.552	**0.434**	56.7	n/a	n/a
Objects	HR reference	n/a	n/a	n/a	68.6	n/a	n/a
Faces	Bicubic	27.38	0.754	0.642	20.4	**0.689**	0.96
Faces	Real-ESRNet	**27.83**	**0.789**	0.453	36.9	0.656	**1.00**
Faces	Real-ESRGAN	26.70	0.762	0.340	56.7	0.608	1.00
Faces	ResShift	27.19	0.755	**0.318**	**60.5**	0.611	1.00
Faces	HR reference	n/a	n/a	n/a	73.6	n/a	n/a
Text	Bicubic	15.09	0.579	0.642	24.2	0.584	n/a
Text	Real-ESRNet	**15.62**	**0.689**	0.409	40.2	**0.574**	n/a
Text	Real-ESRGAN	14.35	0.680	0.215	**52.9**	0.594	n/a
Text	ResShift	14.95	0.668	**0.195**	50.5	0.624	n/a
Text	HR reference	n/a	n/a	n/a	57.4	0.019	n/a

All JPEG levels with intervals: `summary_test.csv`; CER by font size: `summary_text_by_size_test.csv`.

3. Comparison

Paired differences on the same images, method minus Real-ESRNet, 95% intervals [lo, hi]:

Content	JPEG	Method (minus Real-ESRNet)	PSNR (dB)	LPIPS	MUSIQ	CER / ArcFace sim
Objects	none	Real-ESRGAN	-1.89 [-2.06, -1.73]	-0.113 [-0.129, -0.098]	+8.3 [7.0, 9.7]	n/a
Objects	none	ResShift	-0.98 [-1.20, -0.77]	-0.164 [-0.181, -0.148]	+7.6 [6.4, 8.9]	n/a
Faces	none	Real-ESRGAN	-2.43 [-2.56, -2.30]	-0.080 [-0.085, -0.074]	+9.3 [8.6, 10.0]	-0.023 [-0.025, -0.021]
Faces	none	ResShift	-1.36 [-1.52, -1.21]	-0.133 [-0.139, -0.126]	+10.1 [9.3, 10.9]	-0.004 [-0.006, -0.003]
Text	none	Real-ESRGAN	-1.47 [-1.59, -1.34]	-0.077 [-0.095, -0.061]	+5.6 [5.0, 6.3]	+0.120 [0.094, 0.146]
Text	none	ResShift	+0.47 [0.36, 0.58]	-0.098 [-0.119, -0.078]	+4.6 [3.8, 5.5]	+0.023 [0.013, 0.033]
Objects	10	Real-ESRGAN	-1.18 [-1.28, -1.08]	-0.155 [-0.172, -0.138]	+19.8 [18.1, 21.5]	n/a
Objects	10	ResShift	-0.86 [-1.02, -0.72]	-0.161 [-0.181, -0.141]	+16.0 [13.9, 18.1]	n/a
Faces	10	Real-ESRGAN	-1.13 [-1.20, -1.06]	-0.112 [-0.118, -0.106]	+19.8 [18.3, 21.3]	-0.048 [-0.054, -0.041]
Faces	10	ResShift	-0.63 [-0.70, -0.57]	-0.135 [-0.143, -0.127]	+23.6 [22.2, 25.0]	-0.045 [-0.053, -0.036]
Text	10	Real-ESRGAN	-1.27 [-1.33, -1.21]	-0.194 [-0.224, -0.165]	+12.7 [11.7, 13.8]	+0.021 [0.002, 0.040]
Text	10	ResShift	-0.68 [-0.78, -0.58]	-0.214 [-0.252, -0.176]	+10.3 [9.2, 11.5]	+0.051 [0.035, 0.067]

JPEG below 30. Per image, the loss from JPEG 30 to 10 minus the loss from 50 to 30; positive means faster degradation below 30:

Method	Content	PSNR: drop 30 to 10 minus drop 50 to 30 (dB)	SSIM	LPIPS	Verdict
Bicubic	objects	+0.91 [0.83, 0.99]	+0.030 [0.028, 0.032]	+0.038 [0.032, 0.044]	supported
Bicubic	faces	+1.37 [1.31, 1.43]	+0.028 [0.027, 0.029]	+0.059 [0.056, 0.063]	supported
Bicubic	text	+0.50 [0.46, 0.55]	+0.021 [0.019, 0.023]	+0.074 [0.067, 0.081]	supported
Real-ESRNet	objects	+0.91 [0.81, 1.02]	+0.032 [0.029, 0.036]	+0.059 [0.053, 0.065]	supported
Real-ESRNet	faces	+1.38 [1.29, 1.47]	+0.022 [0.021, 0.023]	+0.050 [0.047, 0.053]	supported
Real-ESRNet	text	+1.26 [1.11, 1.40]	+0.072 [0.068, 0.076]	+0.132 [0.119, 0.146]	supported
Real-ESRGAN	objects	+0.73 [0.58, 0.89]	+0.025 [0.019, 0.031]	+0.070 [0.062, 0.080]	supported
Real-ESRGAN	faces	+1.17 [1.05, 1.29]	+0.021 [0.019, 0.022]	+0.064 [0.059, 0.069]	supported
Real-ESRGAN	text	+1.27 [1.09, 1.45]	+0.068 [0.064, 0.072]	+0.064 [0.050, 0.080]	supported
ResShift	objects	+0.69 [0.58, 0.79]	+0.028 [0.022, 0.034]	+0.079 [0.070, 0.090]	supported
ResShift	faces	+1.03 [0.95, 1.10]	+0.019 [0.017, 0.021]	+0.059 [0.054, 0.065]	supported
ResShift	text	+1.11 [0.96, 1.28]	+0.072 [0.068, 0.076]	+0.042 [0.032, 0.051]	supported

Verdicts (`results/analysis/hypothesis_checks_test.json`):

- *Generative models look sharper but lose PSNR:* supported in 11 of 12 cells. Against Real-ESRNet, LPIPS is lower, MUSIQ higher and PSNR lower, all intervals excluding zero. Exception: ResShift on text without JPEG, PSNR +0.47 [0.36, 0.58] dB.
- *They invent letters on text:* supported. CER rises while LPIPS falls: Real-ESRGAN +0.120 [0.094, 0.146], ResShift +0.023 [0.013, 0.033]. For Real-ESRGAN the rise is +0.290 [0.254, 0.327] at 24 px and -0.003 [-0.008, 0.002] at 48 px. In 111 of 384 text inputs, LPIPS ranks a generative method best while its CER is the worst of the 4.
- *They alter identity on faces:* partly supported. ArcFace similarity is lower than for Real-ESRNet: Real-ESRGAN -0.023 [-0.025, -0.021], ResShift -0.004 [-0.006, -0.003] without JPEG; -0.048 [-0.054, -0.041] and -0.045 [-0.053, -0.036] at JPEG 10. Every learned output still matches its HR face at the 0.182 threshold.
- *All models degrade sharply below JPEG 30:* supported in 12 of 12 cells for PSNR, SSIM and LPIPS. The extra PSNR loss is +0.50 to +1.38 dB. At JPEG 10, Real-ESRGAN and ResShift fall below bicubic in mean PSNR on all 3 content types; Real-ESRNet stays above it.

Dev and test agree. The best method is the same on dev and test in 55 of 60 content, JPEG and metric cells (`dev_test_consistency.json`).

4. Runtime and memory

Tesla T4, 3 warm-up runs; time per image excludes loading and saving. Peak memory: `torch.cuda.max_memory_allocated` per method, weights included.

Method	Device	Precision	Mean s/image (test)	Median s/image (test)	Peak GPU memory (MiB)	Wall time dev (s)	Wall time test (s)
Bicubic	CPU (Kaggle)	uint8 (Pillow, CPU)	0.003	0.003	n/a	17	76
Real-ESRNet	Tesla T4	float32	0.156	0.159	208	58	251
Real-ESRGAN	Tesla T4	float32	0.161	0.161	208	67	291
ResShift	Tesla T4	float32 weights, float16 autocast	0.890	0.889	2647	265	1098

Total wall time per notebook: dev 409 s, test 1718 s, seed runs 4327 s. The 4 extra ResShift seeds take 0.873 to 0.883 s per image.

Seed variability. At JPEG none and 10, the ResShift mean over images moves across 5 seeds by at most 0.047 dB PSNR, 0.0035 LPIPS, 1.45 MUSIQ, 0.0029 CER and 0.0058 ArcFace similarity. The per-image PSNR standard deviation averages 0.11 to 0.22 dB. Seed 0 ran on CUDA 12.8, seeds 1 to 4 on CUDA 13.0; the CUDA effect on dev is 0.0035 dB per image (`seed_variability_test.json`).

5. Failure analysis

Cases follow rules fixed before the test run; crops are centred on the largest error of the failing method (Section 7).

(a) LPIPS prefers wrong letters. Rule: LPIPS ranks a generative method best and its CER is the worst; largest CER gap. `text_test_075` (24 px words), JPEG 50, Real-ESRGAN: LPIPS 0.157 (best), CER 0.508 against 0.092 for bicubic. Observation: strokes are sharp, but many glyphs differ from the HR letters; Tesseract reads "circle enough winter" as "circlo Mop wintor". Likely cause: the GAN loss rewards high-contrast strokes, and LPIPS features respond to edges more than to letter identity. Evidence: the CER penalty of Real-ESRGAN peaks at 24 px (+0.290) and vanishes at 48 px. Confirming experiment: a font-size sweep in 2 px steps with the CER gap and the LPIPS-CER rank correlation per size.

(b) Lowest identity similarity. Rule: faces without JPEG, lowest ArcFace similarity of a generative method. `face_00591`, Real-ESRGAN: 0.912 against 0.944 for Real-ESRNet and 0.962 for bicubic; PSNR 27.84 against 30.33 dB. Observation: Real-ESRGAN draws dense, high-contrast hair strands that the input does not contain. Likely cause: GAN texture also reaches the face region and shifts the embedding. Evidence: Real-ESRGAN is the lowest learned method on this image at JPEG none and 50, yet it still matches above the threshold. We cannot exclude that the hair acts through the face alignment. Confirming experiment: interpolate Real-ESRNet and Real-ESRGAN weights and check whether similarity falls with the GAN share; repeat with hair masked.

(c) Generative texture below bicubic. Rule: JPEG 10, largest PSNR deficit of a generative method below bicubic. `object_0880`, ResShift: 21.31 against 24.57 dB (-3.27 dB), SSIM 0.345 against 0.561, while MUSIQ is 63.9 (HR 67.6). Observation: the HR shows fine feathers; ResShift paints a rock-like texture and a dark edge. Likely cause: the diffusion prior fills missing detail with texture that does not align with the true one. Evidence: the

deficit is 2.60 dB without JPEG and 4.56 dB at JPEG 50, so compression is probably not the main cause here. Confirming experiment: relate each object's deficit to the high-frequency energy of its HR image.

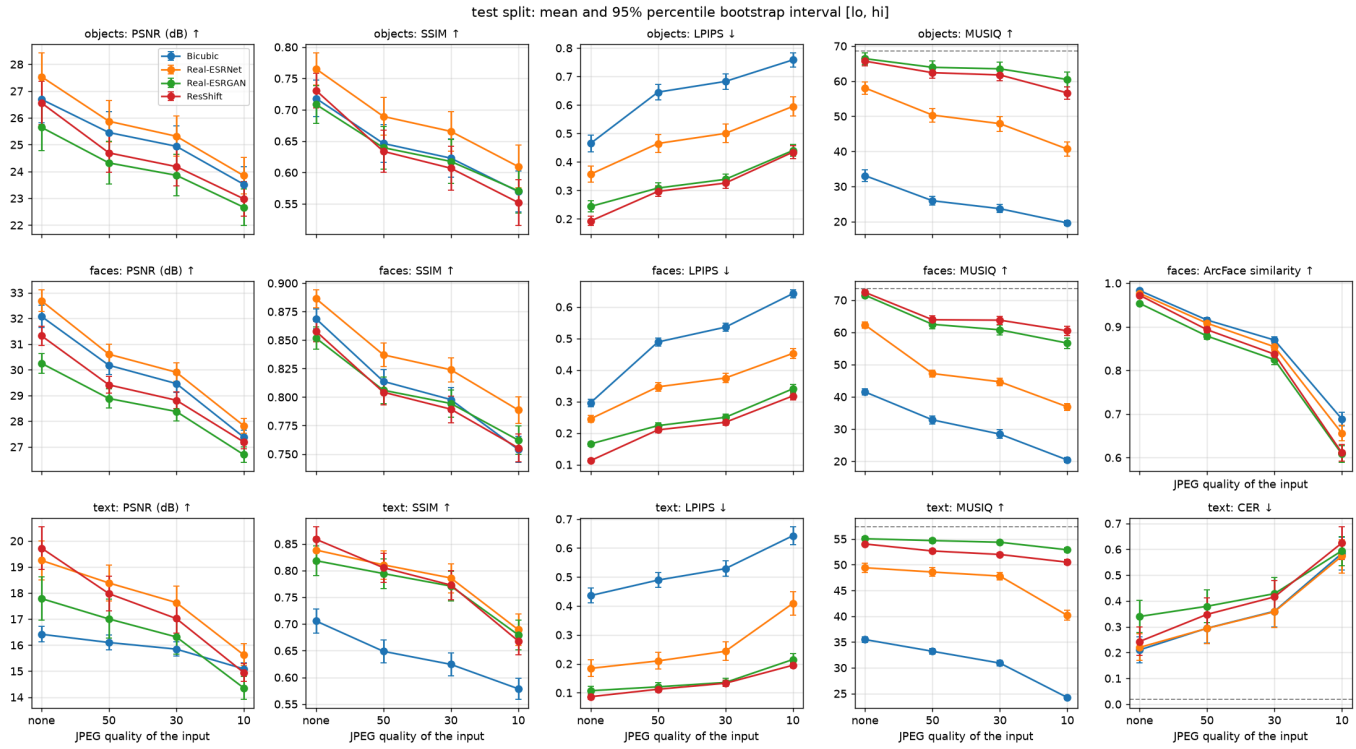
(d) Smallest font with ResShift. Rule: ResShift, 16 px, no JPEG, highest CER. `text_test_069` (random codes): CER 0.751; bicubic 0.764, Real-ESRNet 0.760, Real-ESRGAN 0.811, HR 0.040. ResShift has the best LPIPS here (0.178). Observation: letters are 4 px tall in the input; ResShift draws sharp strokes that form other characters. Likely cause: the input no longer identifies the characters, and codes give no word context. Evidence: at 16 px the mean CER without JPEG is 0.605 to 0.787 for all methods against 0.018 for HR. Confirming experiment: CER against input letter height, words and codes reported separately.

6. Limitations

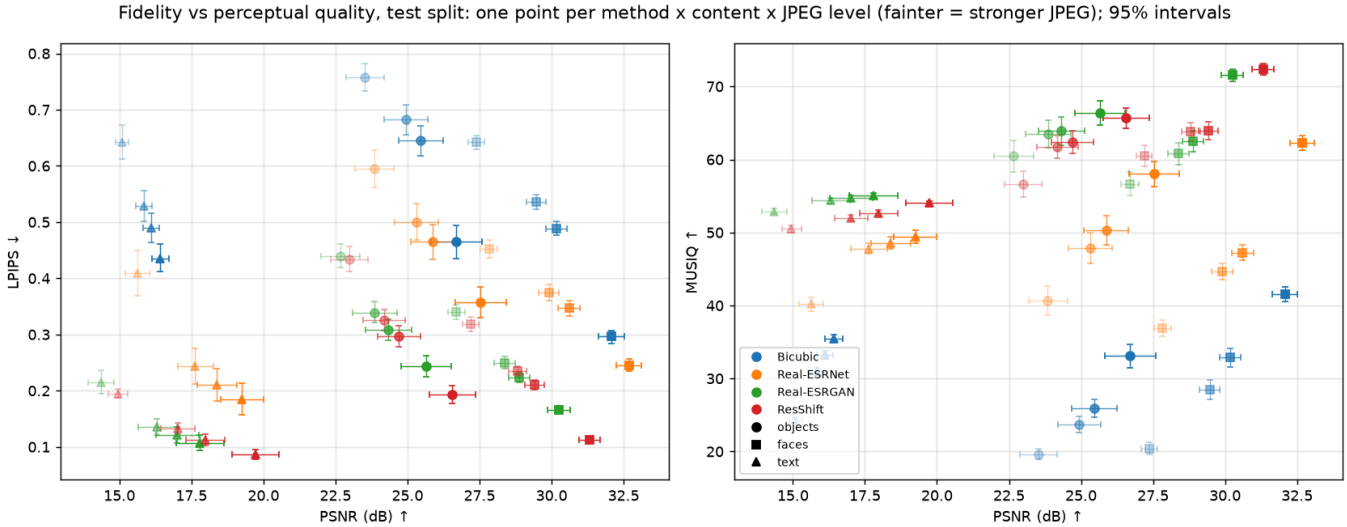
- ResShift differs from RRDBNet in architecture, training data, autoencoder, sampling and precision; gains cannot be attributed to diffusion alone.
- The ResShift v3 configuration lists FFHQ in its training data; overlap with our faces cannot be excluded.
- Intervals are pointwise, without multiple-comparison correction, and the report reads many of them.
- ResShift outputs differ between our 2 GPU environments by up to 0.032 dB per image; the main test run and the seed runs used different CUDA builds.
- The ArcFace threshold rests on 190 impostor pairs. All learned outputs match at it, so only the similarity separates the methods.
- CER uses one OCR engine and configuration. It measures readability for Tesseract, not for people.
- MUSIQ was trained on natural photographs; its scores on rendered text are hard to interpret.
- The test split was accessed once before this stage (`cc9a873`); the final run used frozen settings, but the split was not untouched.
- Failure crops follow the largest pixel error; in case (b) this is the hair, not the face that ArcFace uses.

7. Figures

Metrics against input JPEG quality (test, mean and 95% interval [lo, hi]; dashed: HR reference):

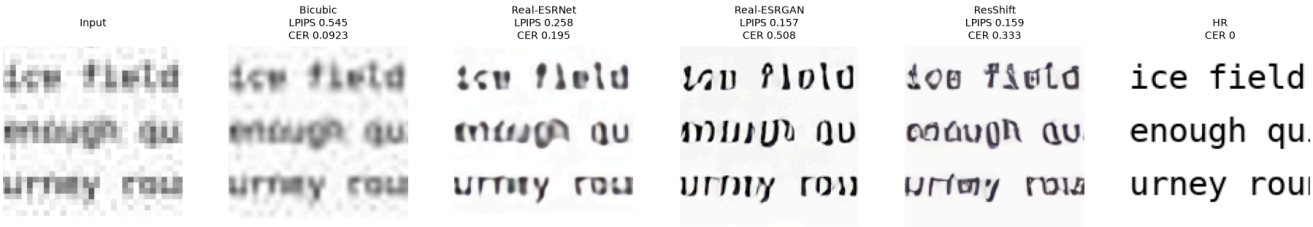


Fidelity against perceptual quality (test): one point per method, content and JPEG level, with 95% intervals.



Failure cases (a) to (d): input, the 4 methods and HR, with the values of the shown image.

Case a: text_test_075, JPEG 50, crop (235, 359, 363, 487)



Case b: face_00591, JPEG none, crop (189, 7, 317, 135)



Case c: object_0880, JPEG 10, crop (0, 148, 128, 276)



Case d: text_test_069, JPEG none, crop (0, 128, 128, 256)

